DEJASHINI T P

AI Full Stack Engineer Intern Candidate | Generative AI · LLM Integrations · React · FastAPI · Python

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EDUCATION

Saveetha Engineering College — B.E. Electronics and Communication Engineering 2024 – 2028
CGPA: 8.74

TECHNICAL SKILLS

Tools: Git/GitHub (Version Control), Postman, Maven, VS Code
Core CS: Data Structures & Algorithms, Object-Oriented Programming (OOP), Software Design Principles
AI / Machine Learning: Machine Learning, Computer Vision, YOLOv8, OpenCV, Random Forest, SHAP
Generative AI / LLMs: LLM & Transformer-based NLP Integration, Speech-to-Text (Whisper), Model Fine-tuning & Optimization
Backend & APIs: Python (FastAPI), RESTful APIs, WebSockets, JWT Authentication
Frontend: React, JavaScript, HTML5, CSS3
Programming Languages: Python, Java, JavaScript, C

EXPERIENCE

REtech Solutions Pvt. Ltd. — Artificial Intelligence Intern Jun 2025 – Jul 2025

- Achieved 88% mAP@50 (89% precision, 86% recall) by training and fine-tuning YOLOv8 on a curated dataset of 2,000+ images across 5 classes, using targeted preprocessing and augmentation.
- Engineered a real-time computer vision inference pipeline in Python (YOLOv8 + OpenCV) that moved detection from prototype to production-ready, cutting manual review effort for the team.

PROJECTS

Forest Carbon AI — Satellite-Based Forest Carbon Stock Estimation *(Python, Google Earth Engine, React, FastAPI)* — github.com/Dejashini123/ForestCarbonAI

- Built an end-to-end AI pipeline estimating forest carbon stock from Sentinel-2 imagery and NASA GEDI biomass data, engineering a 3,126-sample geospatial dataset across 12 remote-sensing features.
- Trained a Random Forest regression model (R^2 = 0.39, MAE = 30.24, RMSE = 43.25) — within the expected range for satellite-based biomass estimation given sensor noise — and deployed it as a full-stack app (React front end + FastAPI RESTful back end) with geospatial visualization and SHAP-based explainability.

AI Smart Traffic & Piezoelectric Energy System *(Python, YOLOv8, OpenCV, IoT)* — github.com/Dejashini123/AI-Smart-Traffic-Piezo-System

- Reduced false detections by 18% by training YOLOv8 on 1,200+ annotated images (91% mAP), applying augmentation and hyperparameter tuning for model optimization.
- Designed and prototyped a piezoelectric energy-harvesting circuit that converts vehicular pressure into measurable electrical output, integrating it with the traffic-detection system for a self-powered smart-traffic concept.

AI-Powered Moderation Engine *(Python, WebSockets, Transformers, Whisper, JWT)* — github.com/Dejashini123/ai-moderation-engine

- Architected a real-time moderation platform handling 500+ concurrent messages at sub-200ms latency using WebSockets for both text and voice streams.
- Automated content moderation by integrating LLM/Transformer-based NLP models and Whisper speech-to-text via model-inference APIs, with JWT authentication securing all client connections.

ACHIEVEMENTS

- Solved 50+ Data Structures & Algorithms problems on LeetCode (actively increasing volume and coverage of medium/hard problems).
- Built an AI-based Smart Traffic Management System (YOLOv8 + IoT) at a 24-hour National Hackathon.
- Competing in Smart India Hackathon (SIH) 2026, developing an AI system to detect anomalies and fraud in MPLADS fund implementation.

CERTIFICATIONS

- AWS Educate — Machine Learning Foundations · IBM — Machine Learning with Python
- IBM — Exploratory Data Analysis for ML · JPMorganChase — SWE Job Simulation (Forage) · L&T EduTech — Microcontroller and Industrial Applications